

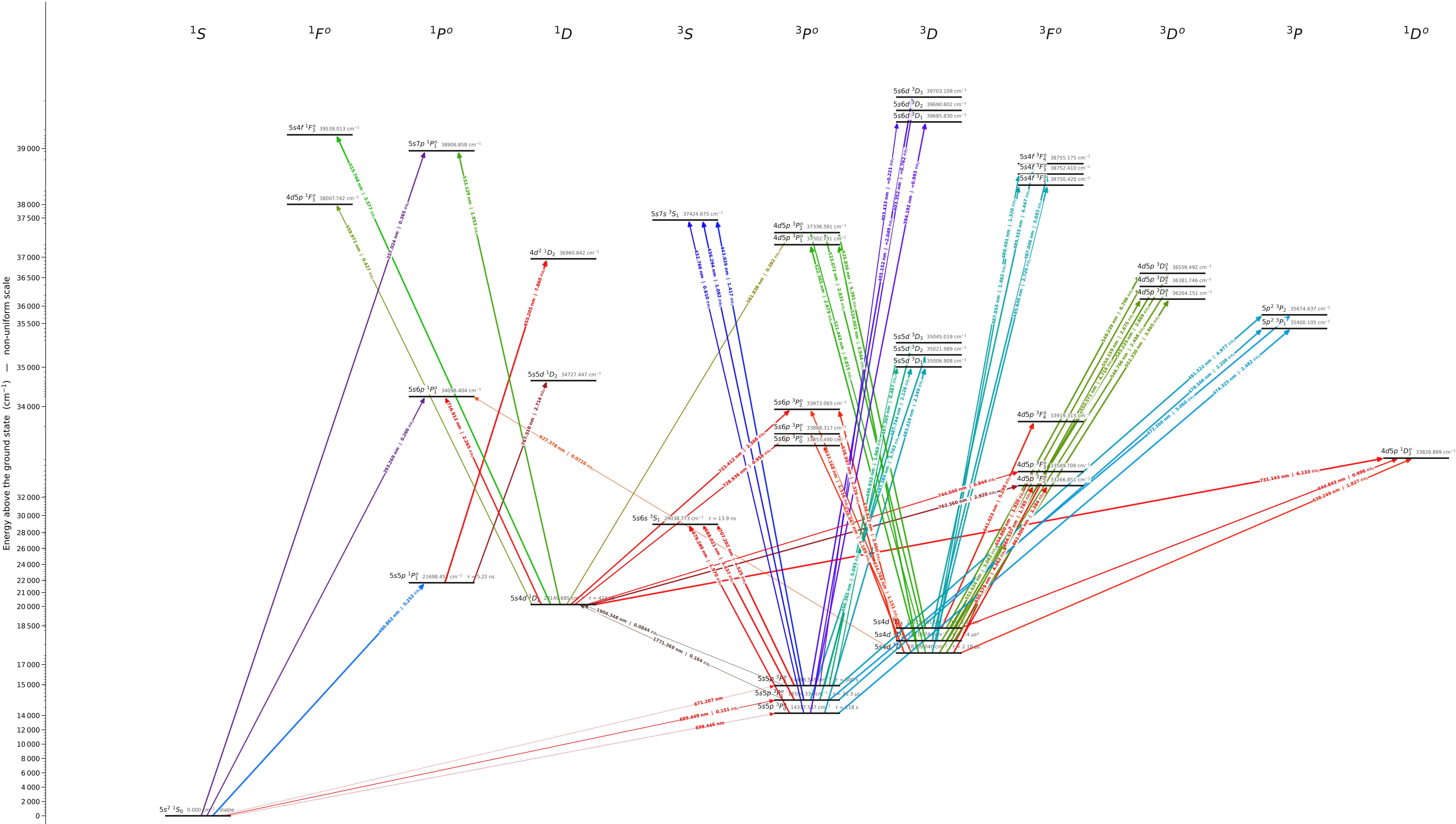
Strontium-84

⁸⁴Sr (Sr I), nuclear spin *I* = 0. Energy levels and transitions below 2 μm: 40 levels, 70 lines.

Sr⁺ ionisation limit 45932.20 cm⁻¹ = 5.69487 eV (217.712 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|(U||er||J)|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ (U er J) $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5s5p ³ P ₀ – 5s6d ³ D ₁	394.1924	394.0808	760.5232	≈0.993	1.088e+07	–	–	394 nm
5s5p ³ P ₂ – 5s6d ³ D ₃	403.1516	403.0377	743.6221	≈2.049	1.854e+07	–	–	403.15 nm
5s5p ³ P ₂ – 5s6d ³ D ₂	403.3517	403.2378	743.2532	≈0.762	3.588e+06	–	–	403.35 nm repumper
5s5p ³ P ₂ – 5s6d ³ D ₁	403.4326	403.3187	743.1041	≈0.221	5.029e+05	–	–	403.43 nm
5s ² ¹ S ₀ – 5s5p ¹ P ₁ ^o	460.8624	460.7333	650.5032	5.252	1.903e+08	30.51	100	461 nm blue MOT / Zeeman slower / imaging
5s5p ³ P ₀ – 5p ² ³ P ₁	474.3248	474.1922	632.0404	2.482	3.900e+07	–	–	474 nm
5s5p ³ P ₀ – 5s5d ³ D ₁	483.3393	483.2042	620.2526	2.349	3.300e+07	–	–	483 nm repumper
5s5p ³ P ₂ – 5s5d ³ D ₃	496.3648	496.2263	603.9761	5.093	6.140e+07	–	–	496.36 nm
5s5p ³ P ₂ – 5s5d ³ D ₂	496.9328	496.7942	603.2857	1.969	1.280e+07	–	–	496.93 nm repumper
5s5p ³ P ₂ – 5s5d ³ D ₁	497.3055	497.1668	602.8336	0.487	1.300e+06	–	–	497.30 nm
5s ² ¹ S ₀ – 5s5p ³ P ₂	671.2065	671.0212	446.6471	–	1.520e-04	3.183e-10	7.6	671 nm magnetic-quadrupole line
5s5p ³ P ₀ – 5s6s ³ S ₁	679.2894	679.1020	441.3324	1.970	8.360e+06	11.43	11.6	679 nm repumper
5s5p ³ P ₂ – 5s6s ³ S ₁	688.0211	687.8313	435.7315	3.437	2.450e+07	11.43	34.1	688 nm
5s ² ¹ S ₀ – 5s5p ³ P ₁ ^o	689.4491	689.2589	434.8290	0.151	4.700e+04	0.007479	100	689 nm red MOT
5s ² ¹ S ₀ – 5s5p ³ P ₀	698.4456	698.2530	429.2281	–	–	1.349e-09	–	698 nm optical clock / optical qubit
5s5p ³ P ₂ – 5s6s ³ S ₁	707.2022	707.0072	423.9134	4.525	3.910e+07	11.43	54.3	707 nm repumper
5s5p ³ P ₁ ^o – 5s4d ¹ D ₂	1771.3690	1770.8854	169.2434	0.164	1.950e+03	0.0003772	82	1D2 decay returning atoms to the ground state via 3P1
5s5p ³ P ₂ – 5s4d ¹ D ₂	1904.3484	1903.8286	157.4252	0.084	4.180e+02	0.0003772	17.7	1D2 decay to the metastable reservoir state

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
5s5p ³ P ₀ – 5s6d ³ D ₁	394.192	88-84	91.05(4)
5s5p ³ P ₂ – 5s6d ³ D ₁	394.192	88-86	54.60(3)
5s5p ³ P ₀ – 5s6d ³ D ₁	394.192	88-87	51.641(28)
5s5p ³ P ₂ – 5s6d ³ D ₃	403.152	88-84	85(10)
5s5p ³ P ₂ – 5s6d ³ D ₃	403.152	88-86	41(10)
5s5p ³ P ₂ – 5s6d ³ D ₃	403.152	88-87	53(5)
5s5p ³ P ₂ – 5s6d ³ D ₂	403.352	88-84	81.0(20)
5s5p ³ P ₂ – 5s6d ³ D ₂	403.352	88-86	49.0(20)
5s5p ³ P ₂ – 5s6d ³ D ₂	403.352	88-87	20(5)
5s5p ³ P ₂ – 5s6d ³ D ₁	403.433	88-84	77(10)
5s5p ³ P ₂ – 5s6d ³ D ₁	403.433	88-86	30(10)
5s5p ³ P ₂ – 5s6d ³ D ₁	403.433	88-87	18.0(20)
5s ² ¹ S ₀ – 5s5p ¹ P ₁ ^o	460.862	84-88	-275(3)
5s ² ¹ S ₀ – 5s5p ¹ P ₁ ^o	460.862	86-88	-126.0(20)
5s ² ¹ S ₀ – 5s5p ¹ P ₁ ^o	460.862	87-88	-47.0(20)
5s5p ³ P ₀ – 5s5d ³ D ₁	483.339	88-84	90.00
5s5p ³ P ₀ – 5s5d ³ D ₁	483.339	88-86	40.00
5s5p ³ P ₀ – 5s5d ³ D ₁	483.339	88-87	32.00
5s5p ³ P ₂ – 5s5d ³ D ₃	496.365	88-84	91.60(20)
5s5p ³ P ₂ – 5s5d ³ D ₃	496.365	88-86	46.60(20)
5s5p ³ P ₂ – 5s5d ³ D ₃	496.365	88-87	27.0(20)
5s5p ³ P ₂ – 5s5d ³ D ₂	496.933	88-84	92.80(20)

Shift = ν(first isotope) – ν(second isotope).